

PRAJAY SANDIPKUMAR MEHTA

+19452416462 | prajaymehta14@gmail.com | [LinkedIn](#)

SUMMARY

- Strong experience in utilizing Python and JAVA in data related and backend development of software applications.
- Proficient in applying Python, JAVA, SQL/NoSQL, Flask, cloud technologies (AWS) and machine learning models to help in all phases of the software development life cycle and agile projects.
- Adept at crafting visually engaging and informative data visualizations through tools like Python's Matplotlib and Seaborn, as well as Tableau, enhancing the accessibility of complex data insights for stakeholders across the organization.
- Completing my AWS Solutions Architect certification to further strengthen my cloud expertise and am eager to apply these skills and credentials in real-world projects.

SKILLS

Scripting Languages: Python, JAVA, Spring Boot, R, Scala, MATLAB

Database: Relational(MySQL, Postgres), NoSQL(MongoDB), Cloud(AWS EC2, S3, ELB, RDS, ECS, Route 53, etc.)

Big Data: Hadoop, Apache Spark, MLlib

Data Analysis and Visualization: Hypothesis Testing, A/B testing, SQL, Excel, Tableau, R, Pandas, NumPy, Matplotlib

Testing and QA: Jest, React Testing Library, Selenium, Chai, Unit Testing, Integration Testing,

DevOps and Cloud Tools: Docker, AWS, Git, Microservices Architecture, DataBricks, Jupyter, Tableau, JIRA, ETL, VS Code

EDUCATION

Master of Science, Computer Science | University of Texas at Dallas, Richardson TX

Bachelor of Engineering, Computer Engineering | Gujarat Technological University, India

PUBLICATIONS

- [Human Machine Comparison for Cross Race face verification.](#)
- [Designing Cross-Race tests for Forensic Facial Examiners, Super-recognizers, and Face Recognition Algorithms.](#)

PROFESSIONAL EXPERIENCE

University of Texas at Dallas.

Research Assistant.

Richardson, TX, USA

February 2023 - January 2024

- Employed advanced machine learning and deep learning techniques, including neural network architectures such as ArcFace, DeepFace, and Facenet, to analyze the NIST dataset for Facial Recognition.
- Communicated data-driven insights to executive teams, leading to a 20% increase in data-informed decision-making.
- Optimized pipeline for algorithms to enhance processing speed and recognition of non-frontal facial images, improving model accuracy by 12%.
- Automated feature extraction process for efficient data analysis and model training, thus improving response time by 2 hours.
- Applied advanced statistical techniques using R and Python to analyze and interpret results, ensuring robust insights from experimental comparisons between machine and human recognition abilities using PsychoPy and Pavlovia.

Comnet Development India

Jr. Data Scientist

Surat, Gujarat, India

January 2022 - April 2022

- Utilized cloud for real-time data storage and processing, ensuring accurate inventory levels are maintained across all warehouses and sales channels.
- Cleaned and transformed raw inventory data using Python (Pandas, NumPy) to resolve inconsistencies, handle missing values, and prepare data for analysis and modeling.
- Contributed to the development of predictive models for demand forecasting and inventory optimization by assisting with feature engineering and data preparation which reduced operational costs by 20 %.
- Provided actionable business insights via interactive visualizations made using Tableau, aiding in strategic decision-making.
- Developed a responsive web page (Admin Panel) using Angular, JavaScript to monitor and manage real-time inventory data, improving user experience for warehouse.

Hands Together Tech

Jr. Software Development Engineer

Surat, Gujarat, India

March 2021 - July 2021

- Contributed to the development of a caregiving platform featuring a mobile application (React + Native) and a responsive web-based admin panel for multi-level administrators.
- Focused on the admin panel, building and enhancing UI components (polls, news feed, alerts) using React and Material UI to improve user experience for branch and central admins.
- Integrated REST APIs, implemented front-end validations, and wrote unit tests to ensure robust and maintainable code.
- Translated wireframes into production-ready UI components, enhancing usability for healthcare professionals through iterative feedback loops.

PROJECTS

Credit Card Fraud Detection:

Developed a data exploration and analysis pipeline for a credit card transaction dataset using **Python**. Implemented data preprocessing, visualization, and built a predictive model utilizing **Keras, Scikit-learn, NN with SMOTE** to detect fraudulent transactions. Employed libraries such as **pandas, numpy, and matplotlib** to facilitate data manipulation and visualization.

TaskTap MVP: ([TaskTap Landing Page](#)) React, TypeScript, Tailwind CSS, Supabase, Git

Building TaskTap, a lightweight task-based jobs marketplace MVP leveraging cloud-native architecture to match users with quick gigs.

Designed for scalability and real-time task management using modern web technologies.